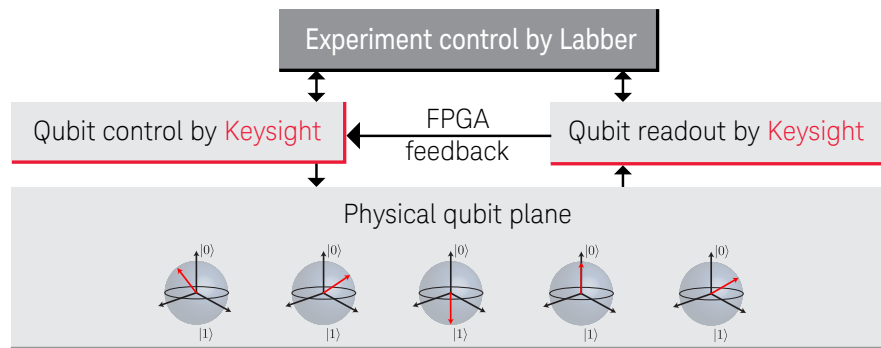


Keysight Technologies

Quantum Researchers Toolkit + Labber

S5060A Solution Brief



Researchers in quantum information systems are focused on identifying and improving the performance of single quantum bit (qubit) and multi-qubit architectures. This research requires the investigation of key metrics such as quantum coherence, cross-talk, qubit-gate and readout fidelities.

Keysight Technologies and Lab Control Scandinavia AB have developed a powerful control system for use in single and multi-qubit research lab experiments. This solution enables evolution to multiple qubit experiments and provides a rich development environment for experiments in pulse control, pulse shaping, signal demodulation, and feedback, as used in qubit labs.

Researchers have several architectural path options for qubit control experiments. However, it is important to consider the short and long term factors associated with each path when selecting a given architecture path. One way could be to use multiple individual instrumentation components from multiple suppliers, and to design and build a customized control system. Another would be to use an integrated control solution, such as the Keysight Researchers Toolkit.

Individual instruments are often used for small scale qubit experiments. However, this approach lacks the scalability necessary to move from single qubit to multi-qubit experiments. The limited ability to control the instruments may constrain the types of experiment that can be performed. For example, even if the instruments can be synchronized for signal phase coherence, multiple instruments may not be able to support the ultra-low

latency, decision/control feedback loops that are needed for quantum error correction. This architecture can become prohibitively expensive as the experiment scales to multi-qubit operations. In addition, the extra physical space occupied by the instruments can become a problem in limited lab environments.

A custom designed control system enables a solution optimized to the current experiment. However, over time it may lack the flexibility required to adapt to different paths of research as the experiments evolve. This limitation then results in the need for enhancements or even a new generation of customized solution. This custom design and re-design overhead can become an expensive drain on limited resources.

Keysight offers an integrated control system that is built on an industry-standard architecture using PXIe chassis and modules, yet retains the flexibility to perform in both low and high scale configurations, enables user programming and can adapt to evolving research requirements.

Keysight Technologies can help you take the lead - from single to multi qubit control using a flexible, open control architecture. Gain the advantage from Keysight's ongoing collaboration in quantum research labs. This expertise is swiftly embodied into Keysight quantum control solutions, tuned for the specific demands of quantum research.



Unlocking Measurement Insights



Multi-Qubit Control Challenges

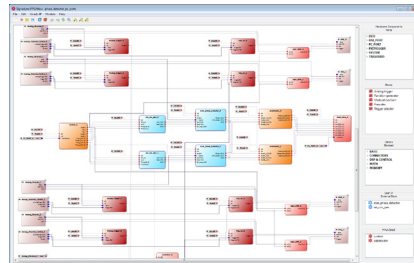
As qubit experiments scale from single to multi-qubit, the control system needs to grow the number of control ports and readout inputs while maintaining the tightly synchronized waveform coherence across multiple ports.

NEED

Open programmable architecture to enable efficient researcher development.

CHALLENGE

Enabling researchers to have an open development environment for both, control and signal path, while maintaining critical run-time performance.



www.keysight.com/find/m3602a

SOLUTION

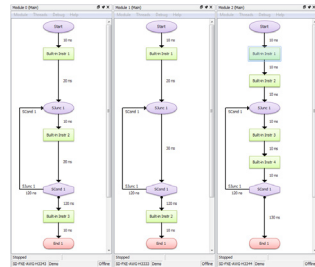
Keysight provides a flexible environment for development of the necessary control sequences and for signal path manipulation (e.g., for pulse shaping and phase detection in the module FPGAs). This environment enables rapid integration of logic blocks into the signal path and complements the Application software APIs and “Hard Virtual Instrumentation” (HVI) controls.

NEED

Coordinated multi-port control with tight amplitude, phase and timing requirements for signals.

CHALLENGE

Maintaining phase coherence across multiple ports and across multiple chassis.



www.keysight.com/find/m3601a

SOLUTION

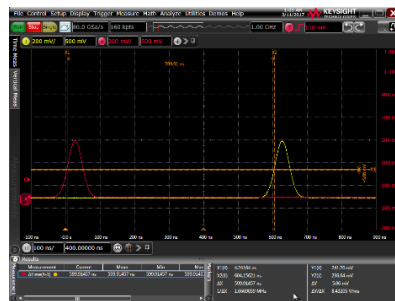
Keysight HVI solution provides high level sequence programming and enables synchronized phase coherent signal control across multiple ports, multiple modules and even across multiple chassis.

NEED

Fast feedback loops to control and correct qubit state.

CHALLENGE

Providing a high-speed chain of: signal demodulation, control decision and synchronized pulse generation in < 600 ns.



SOLUTION

Keysight's control system enables feedback loops to be programmed by the researcher and can achieve pulse output latencies in under 150 ns.

NEED

Cost and space efficient scaling to support the numbers of channels to control multi-qubit experiments.

CHALLENGE

Providing a high-performance architecture that is based on industry standards and can scale up to hundreds of channels.



www.keysight.com/find/modular

SOLUTION

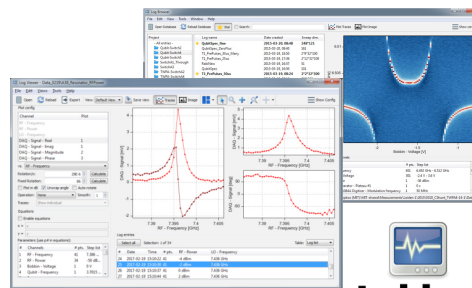
Keysight modular solutions scale easily to a multi-channel architecture while minimizing the total cost of provisioning instrumentation around each qubit. The solution provides up/down conversion to the most commonly used microwave frequencies. This modular and multi-chassis architecture addresses today's scale needs while maintaining a path for future requirements.

NEED

High level control of complex experiments with visualization and results management.

CHALLENGE

Providing an application that can both control the experimental parameters and collect live results data by integrating with multiple hardware components.



www.labber.org

Labber

SOLUTION

Labber software, from Keysight's solution partner, provides experiment control, a qubit fidelity calibration capability and manages experiment result data for analysis and comparison.

Solutions for qubit characterization and experiment control

Keysight Technologies is uniquely positioned to help you improve your time-to-insight while keeping your total cost of qubit research under control. Our quantum research solutions are highly scalable with comprehensive control and measurement capability built on an extensible modular architecture.

Why a Researcher's Toolkit?

Different research groups are focusing on specific qubit implementation technologies and multi-qubit architectures. Between these different approaches the needs for the control system will vary. However, there are common needs shared by most groups. This led to the creation of a Researcher's Toolkit that can be readily adapted to meet researcher needs.

Key attributes for the Toolkit are:

- Phase coherent AWG operations
- High integrity waveform generation
- Rapid digitizer response and performance
- User definable functionality through open development environments
- Compact and cost effective architecture for lower price per qubit control

Supporting Experimenter Development

Keysight's solution provides FPGA based signal processing together with the precision synchronized pulses for qubit state control and readout, plus digitizer architecture for demodulating the qubit output.

Our HVI environment allows researchers to program sequences for each module, enabling synchronized operations on both AWGs and digitizers.

The FPGA logic blocks can be integrated into the signal path for waveform manipulation and connected to the digitizer output path for low latency feedback/error correction decisions.

The modules operate at baseband with up/down conversion to the most commonly used microwave qubit control frequencies up to 20 GHz. The signal conditioning can be accomplished in different domains – analog, digital or a hybrid of the two.

Supporting scalability

With the progression of multi-qubit experiments, characterization of qubit performance becomes increasingly complex. Multi-channel measurement solutions based on traditional bench-top instrumentation are no longer practical or efficient, and new methods are required to ensure reliable and repeatable characterization right to the measurement plane.

Keysight solutions scale to the large port-count required for multi-qubit control and readout, incorporating novel techniques to provide synchronization accuracy and experiment repeatability. A range of modular components make it easy to scale to the number of qubits being characterized.

Experiment Control with Labber

As a solution partner for Keysight, Lab Control Scandinavia AB has enabled full control and integration of Keysight's latest PXIe hardware for signal generation and detection.

Labber is a powerful yet user friendly software application for instrument control and measurement automation. In addition to controlling and automating laboratory measurements, Labber is designed to solve more demanding tasks such as waveform creation, measurement analysis and feedback on the fly, and synchronous operation of multiple instruments. These features will play an important role when performing tasks such as quantum error correction and advanced calibration of multiple-qubit architectures.

For more information on Labber, to review documentation and request trial licenses, visit www.labber.org.

Example Configurations

Minimal – for small scale experiments

- PXIe M9010A 10 Slot Chassis and M9037A Controller
- M3202A AWG, M3102A Digitizer

Scalable – for multi-chassis expansion

- Host PC and M9049A PCIe Host adapter
- M9019A 18 Slot PXIe Chassis and M9023A System Module
- M3202A AWGs, M3102A Digitizers

Improve your qubit research efficiency and time to results

Move fast through qubit characterization experiments: Leverage the researcher's toolkit and Labber as a control solution for your qubit experiments.

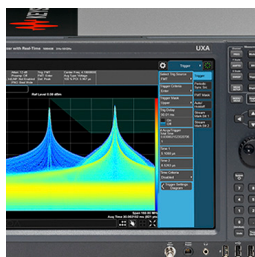
- Flexible, intuitive control and measurement
- Purchase only the functionality you need now
- Expand or upgrade later, as channel requirements grow with experiment scale

At Keysight we continuously develop our test methods to offer solutions that carefully balance cost, scale and accuracy. In quantum research, our solutions help you transform experimental ideas into reality. Keysight's industry-leading solutions provide meaningful results, fast. Contact us to learn how we can help you lead in quantum information systems.

Evolving Since 1939

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Get the best of both worlds: Keysight's measurement expertise and product breadth, combined with channel partner convenience.

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For more information on Keysight Technologies' products, applications or services, please contact your local Keysight office. The complete list is available at:
www.keysight.com/find/contactus

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Unlocking Measurement Insights

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